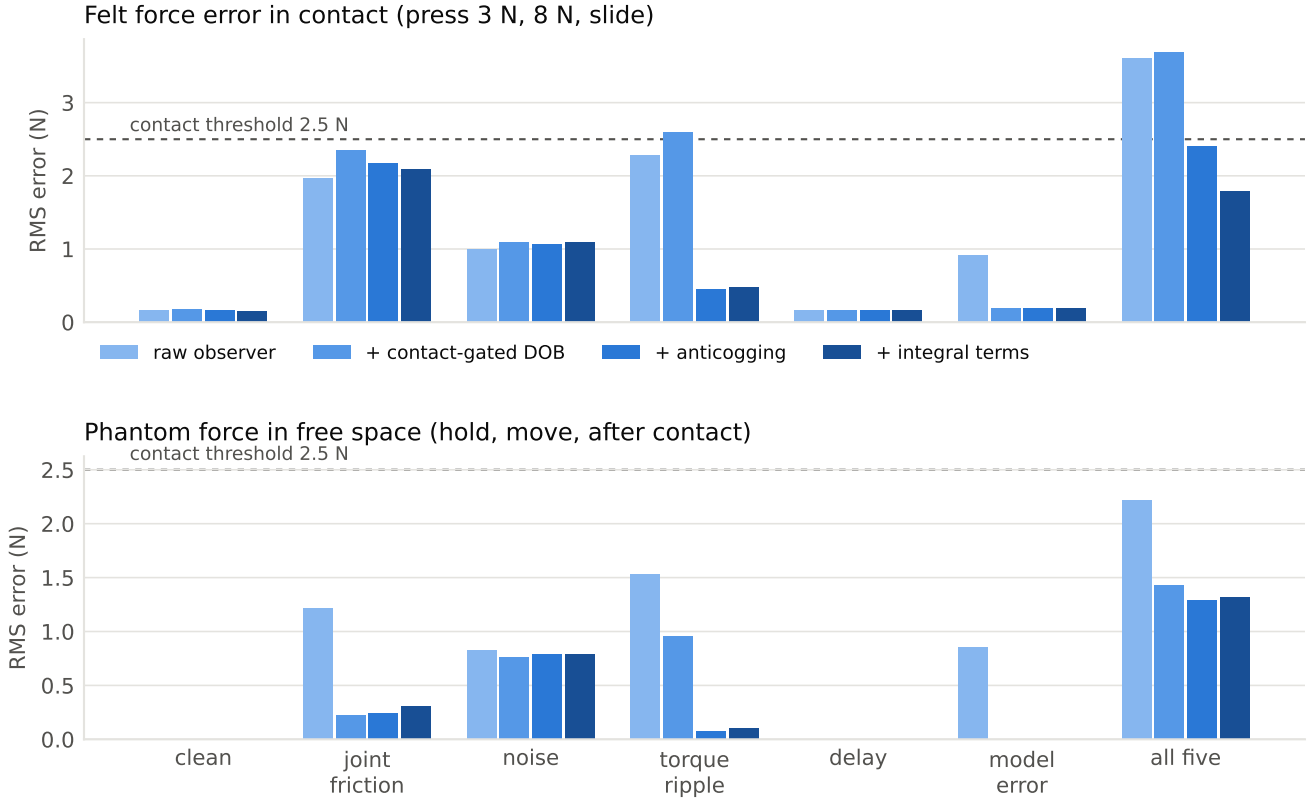
Felt force error in contact (press 3 N, 8 N, slide)
RMS error (N)
0
1
2
3
contact threshold 2.5 N
raw observer
+ contact-gated DOB
+ anticogging
+ integral terms
Phantom force in free space (hold, move, after contact)
RMS error (N)
0.0
0.5
1.0
1.5
2.0
2.5
contact threshold 2.5 N
clean
joint friction
noise
torque ripple
delay
model error
all five